

PRACTICE EXAM

Difficulty: MEDIUM

Questions: 10

Computer Engineering Exam

Section 1: Multiple Choice (4 points each)

Instructions: Choose the best answer for each question.

Question 1: What was one of the main objectives of the "Introduction Lab" conducted on January 22, 2025?

- A) To design a new type of capacitor.
- B) To determine the combined RMS output voltage value of a circuit.
- C) To learn how to use a breadboard.
- D) To study the properties of different signal generators.

Question 2: According to the text, what is a key difference between PyQt5 and PyQt6?

- A) PyQt5 is built on the QT6 library, while PyQt6 is built on the QT5 library.
- B) PyQt6 is more modern and built on the QT6 library for C++.
- C) PyQt5 supports more modules than PyQt6.
- D) There is no difference; PyQt5 and PyQt6 are the same.

Question 3: Which PyQt6 module contains the QMainWindow class?

- A) QtCore
- B) QtGui
- C) QtWidgets
- D) QSplitter

Question 4: According to the Syed Surkhi cover letter, what experience does Syed have with Raspberry Pi?

- A) Building a real-time control system for a rover.
- B) Developing a new operating system for Raspberry Pi.
- C) Designing a Raspberry Pi based mobile phone.
- D) No experience is listed.

Section 2: Short Answer (6 points each)

Instructions: Answer each question in 2-3 sentences.

Question 5: Describe the purpose of the Appendix in the "Introduction Lab" report.

Question 6: Briefly explain the purpose of a QSplitter in PyQt6 and provide an example scenario where it would be useful.

Question 7: Summarize Syed Surkhi's experience with data workflows, as highlighted in their cover letter.

Section 3: Problem-Solving (10 points each)

Instructions: Provide detailed answers, showing your understanding and application of the material.

Question 8: In the context of the "Introduction Lab," explain how the circuit's behavior changes when the capacitor acts as an open circuit versus when it is shorted (refer to the figures mentioned in the text).

Question 9: Suppose you are tasked with recreating the current feed title aspect (aspect ratio wrapper, overlay container, name label, etc.) in PyQt. Describe the steps and PyQt components you would use to achieve this, based on the information provided in the text.

Question 10: Based on the information in Syed Surkhi's cover letter, explain how their previous experiences make them a suitable candidate for the Embedded Developer Intern, Video Products role at Geotab. Focus on specific skills and projects mentioned in the letter that align with the requirements of the position.